

IMAGE PROCESSING USING MATLAB:

Basic Operations



Dr Anil Kumar Maini
is former director,
Laser Science and
Technology Centre,
a premier laser and
optoelectronics
R&D laboratory of
DRDO of Ministry
of Defence



Varsha Agrawal is a
senior scientist with
Laser Science and
Technology Centre
(LASTEC), a premier
R&D lab of DRDO

In this series of four articles, fundamentals as well as advanced topics of image processing using MATLAB are discussed. The articles cover basic to advanced functions of MATLAB's image processing toolbox (IPT) and their effects on different images. Part 1 in this series gives a brief introduction to digital images and MATLAB followed by basic image processing operations in MATLAB including image reading, display and storage back into the disk.

Image processing

Image processing is the technique to convert an image into digital format and perform operations on it to get an enhanced image or extract some useful information from it. Changes that take place in images are usually performed automatically and rely on carefully designed algorithms.

Image processing is a multidisciplinary field, with contributions from different branches of science including mathematics, physics, optical and electrical engineering. Moreover, it overlaps with other areas such as pattern recognition, machine learning, artificial intelligence and human vision research. Different steps involved in image processing include importing the image with an optical scanner or from a digital camera, analysing and manipulating the image (data compression, image enhancement and filtering), and generating the desired output image.

The need to extract information from images and interpret their content has been the driving factor in the development of image processing. Image processing finds use in numerous sectors, including medicine, industry, military, consumer electronics and so on.

In medicine, it is used for diagnostic imaging modalities such as digital radiography, positron emission tomography (PET), computerised axial tomography

(CAT), magnetic resonance imaging (MRI) and functional magnetic resonance imaging (fMRI). Industrial applications include manufacturing systems such as safety systems, quality control and automated guided vehicle control.

Complex image processing algorithms are used in applications ranging from detection of soldiers or vehicles, to missile guidance and object recognition and reconnaissance. Biometric techniques including fingerprinting, face, iris and hand recognition are being used extensively in law enforcement and security.

Digital cameras and camcorders, high-definition TVs, monitors, DVD players, personal video recorders and cell phones are popular consumer electronics items using image processing.

MATLAB

MATLAB, an abbreviation for 'matrix laboratory,' is a platform for solving mathematical and scientific problems. It is a proprietary programming language developed by MathWorks, allowing matrix manipulations, functions and data plotting, algorithm implementation, user interface creation and interfacing with programs written in programming languages like C, C++, Java and so on.

In MATLAB, the IPT is a collection of functions that extends the capability of the MATLAB numeric computing environment. It provides a comprehensive set of reference-standard algorithms and workflow applications for image processing, analysis, visualisation and algorithm development.

It can be used to perform image segmentation, image enhancement, noise reduction, geometric transformations, image registration and 3D image processing operations. Many of the IPT functions support C/C++ code generation for desktop prototyping and embedded vision system deployment.